Submission LINALG1-MID-SUB06

Question LINALG1-MID-Q01

notation changes halfway:

1. det $\approx 2(3)–4(2) \approx -2$
2. Eliminate y, yielding $x \approx 1$; substitute $\Rightarrow y \approx 2$
3. checks: $2(1)+4(2) \approx 10$, $2(1)+3(2) \approx 8$

marks, subscripts, and units are mixed up

result written as: x=1, y=2 (please interpret)

Question LINALG1-MID-Q02

notation changes halfway:

1. $\det(A) \approx 6\cdot4–2\cdot2 \approx 20$
2. $\det(A) \neq 0$, so A is invertible.

marks, subscripts, and units are mixed up

result written as: det=20; invertible (please interpret)

Question LINALG1-MID-Q03

notation changes halfway:

1. $T(1,–1) \approx (1–2,3+1) \approx (–1,4)$
2. Matrix [[1,2],[3,–1]]
3. Both coordinate rules have no constant term, so T is linear.

marks, subscripts, and units are mixed up

result written as: T(1,-1)=(-1,4); matrix [[1,2],[3,-1]]; linear (please interpret)

Question LINALG1-MID-Q04

notation changes halfway:

1. Set c■(1,0)+c■(0,1) ≈ (0,0).
2. Coordinates give c■ ≈ 0 and c■ ≈ 0 only, so the set is independent.

marks, subscripts, and units are mixed up

result written as: only trivial combination; {(1,0),(0,1)} is linearly independent (please interpret)

Question LINALG1-MID-Q05

notation changes halfway:

1. Null(A) is the set of vectors sent to the zero vector.
2. It is the solution space of $Ax \approx 0$ and is a subspace.

marks, subscripts, and units are mixed up

result written as: the vectors mapped to zero; exactly the solution space of Ax=0 (please interpret)

Question LINALG1-MID-Q06

notation changes halfway:

1. det $\approx 5(5)–2(2) \approx 21$
2. Eliminate y, yielding $x \approx 1$; substitute $\Rightarrow y \approx 2$
3. checks: $5(1)+2(2) \approx 9$, $2(1)+5(2) \approx 12$

marks, subscripts, and units are mixed up
result written as: x=1, y=2 (please interpret)

Question LINALG1-MID-Q07
notation changes halfway:
1. $\det(A) \approx 2\cdot 2 - 2\cdot 2 \approx 0$
2. $\det(A) \approx 0$, so A is not invertible.
marks, subscripts, and units are mixed up
result written as: det=0; not invertible (please interpret)

Question LINALG1-MID-Q08
notation changes halfway:
1. $T(1,-1) \approx (1-2,3+1) \approx (-1,4)$
2. Matrix [[1,2],[3,–1]]
3. Both coordinate rules have no constant term, so T is linear.
marks, subscripts, and units are mixed up
result written as: T(1,-1)=(-1,4); matrix [[1,2],[3,-1]]; linear (please interpret)